Algebraic Geometry WS 26/27

RECAP Sheet 3: 28.10-3.11

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R — Recall

1. What is the spectrum of a ring ? What is the map induced on spectra by a ring morphism ?
2. For a ring R, why is $\mathrm{Spec}(R)$ quasi-compact ?
3. For a ring R and a principal open $U \subset \mathrm{Spec}(R)$, why is $\mathcal{O}(U)$ well defined ?
4. For an element $f \in A$, what is the relation between $\mathrm{Spec}(A/f)$ and $V(f)$? Between $\mathrm{Spec}(A[\frac{1}{f}])$ and $D(f)$?

E — Explore

1. What are some examples of rings not of the form $k[X_1, ..., X_n]/I$ for a field k and an ideal I ? Give your own examples with residue fields of different characteristic.
2. In $k[x_1, ..., x_n]$, what are some examples of prime ideals that are not maximal ? Find examples other than the ones in the lecture notes.
3. Describe $\mathrm{Spec}(\mathbb{Z})$.
4. Describe the map $\mathrm{Spec}(f)$ induced by the ring morphism $k[x] \to k[[x]]$. Is it open, closed ?
5. Suppose $f : A \to B$ is a ring morphism such that $\mathrm{Spec}(f)$ is a homeomorphism. Is f an isomorphism ?

C — Connect

1. What is the relation between the notion of an irreducible component and that of a generic point ?
2. What does Hilbert's Nullstellensatz say about $\mathrm{Spec}(k[x_1, ..., x_n]/I)$, for I an ideal ?
3. How are the maximal spectrum and the spectrum of a ring related ? What does that have to do with "functoriality" ?
4. How does Hilbert's Nullstellensatz relate to the statement that for any ring A, closed subsets of $\mathrm{Spec}(A)$ are in one to one correspondence with radicals of A ? Does one imply the other ?

A — Apply

1. Let A be a reduced commutative ring and $X = \mathrm{Spec}(A)$. Suppose X has a generic point x, and suppose $f, g \in A$ are such that $f(x) = g(x) \in \kappa(x)$. Prove that $f = g$. What if x is

A — Apply (continued)

the generic point of a closed subset $V \subset X$?

2. Let $f \in \mathbb{C}[x, y]$. Prove that closed points of $\mathrm{Spec}(\mathbb{C}[x, y]/f)$ are in bijection with complex solutions of the equation $f(x, y) = 0$. What about closed points of $\mathrm{Spec}(\mathbb{R}[x, y]/f)$, if $f \in \mathbb{R}[x, y]$?

3. Prove or disprove the following two statements, where A is a commutative ring:

   - If $f, g \in A$ are such that for all $x \in \mathrm{Spec}(A)$, $f(x) = g(x) \in \kappa(x)$, then $f = g$;
   - If $f \in A$ is such that $f(x) \in \kappa(x)$ is nonzero for all $x \in \mathrm{Spec}(A)$, then f is invertible.

P — Present

Possible starting questions for the tutorial:

- Why do we care about prime ideals rather than maximal ideals ?
- What is different about closed points of $\mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n]/f)$ and closed points of $\mathrm{Spec}(\mathbb{C}[x_1, ..., x_n]/f)$? What do they correspond to intuitively ?
- Does the topological space $\mathrm{Spec}(A)$ know everything about A ?

The tutor may also ask about other central definitions, examples, arguments, or connections from this sheet.